

JONATHAN BUWEMBO

+1 (978) 641-0870 | jkbuwembo@wpi.edu | <https://github.com/JonBuwembo> | <https://www.linkedin.com/in/jonathan-buwembo/>

EDUCATION:

Worcester Polytechnic Institute (WPI), Worcester, MA, Aug 2022 – May 2026

Bachelor of Science in Computer Science, GPA 3.79/ 4.00

Related Courses: Software Engineering, Algorithms, Operating Systems, Embedded Systems, Machine Organization and Assembly, Unified Robotics.

SKILLS:

- **Database Management:** PostgreSQL, SQLAlchemy
- **Frameworks/Libraries:** React.js, Bootstrap, Flask
- **Languages:** Java, C, C++, Python, JavaScript, HTML/CSS, Kotlin
- **Developer Tools:** Git, Linux, VS Code, IntelliJ IDEA

EXPERIENCE:

Undergraduate Research Intern — *Distributed Systems*, WPI June 2023 – Aug 2023

- Built a distributed Hadoop/MapReduce system on a Raspberry Pi cluster to evaluate scalability and parallel data processing.
- Developed performance analysis tools to monitor CPU usage, resource allocation, and efficiency, enabling data-driven optimization decisions.
- Increased system throughput by 55% by improving task scheduling, reducing idle time, and optimizing resource utilization across nodes

PROJECTS:

Software Engineer (Team Project) – *Student Mental Health Application* Sept 2025 – Dec 2025

- Developed a full-stack Android application supporting student mental health at UMass Amherst and WPI, sharing core logic across iOS and Android using Kotlin Multiplatform.
- Built secure APIs to collect and process audio and physiological data for personalized mental health insights.
- Integrated ML models to recommend tailored mental health resources, enabling scalable personalization for future users.

CNN-Based Fruit Image Classification Model – *Deep Learning* Aug 2025 – Present

- Designed and implemented a Convolutional Neural Network (CNN) to classify hundreds of fruit images
- Applied image augmentation techniques to improve model generalization and reduce overfitting.
- Trained the model using the Adam optimization algorithm for efficient convergence, achieving 95% test accuracy.

Full-Stack Developer (Team Project) – *Research Enrollment Application* Mar 2024 – May 2024

- Developed a full-stack web app (Flask, SQLAlchemy, Bootstrap) to streamline WPI research enrollment.
- Implemented secure authentication and skill-based rating system, improving functionality by 20%.
- Optimized database performance and validated reliability with SQLAlchemy and JUnit testing.

Facial Emotion Recognition (ML) – *Machine Learning* Jan 2024 – Mar 2024

- Built a CNN in PyTorch to classify 50,000+ facial images, achieving 55% validation accuracy.
- Implemented data preprocessing and augmentation using PyTorch transforms and NumPy.
- Improved training efficiency with batch normalization and learning rate scheduling.

ACTIVITIES AND LEADERSHIP:

Financial Officer, Dough Club (Social Justice Org) Aug 2024 – Present

- Lead financial planning and fundraising efforts for a student org addressing campus food insecurity.
- Organized donation drives that raised \$1,000+ to support student meal assistance programs.
- Collaborated with campus dining services to allocate funds effectively and maximize impact.

Member, Color Stack June 2025 – Present

- Engaged with a community of underrepresented students in tech through professional development workshops.
- Participated in networking events and career-building programs focused on CS/tech industries.

Member, National Society of Black Engineers (NSBE) Aug 2022 – Present